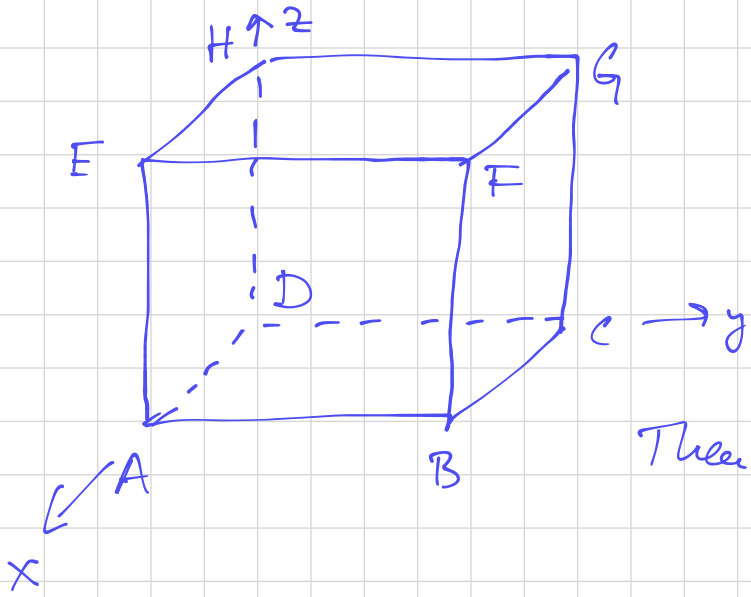


Let point D be the origin: $D = (0,0,0)$

Let the side of the cube to be of length 1.

Then $F = (1,1,1) \Rightarrow \vec{DF} = (1,1,1)$

$B = (1,1,0)$, $H = (0,0,1)$

$\Rightarrow \vec{BH} = (-1,-1,1)$

Therefore $\vec{DF} \cdot \vec{BH} = (1,1,1)\cdot(-1,-1,1) = -1$

On the other hand $\vec{DF} \cdot \vec{BH} = \underbrace{|\vec{DF}|}_{\sqrt{3}}\,\underbrace{|\vec{BH}|}_{\sqrt{3}} \cos\theta_1$

$$\Rightarrow \cos\theta_1 = -\frac{1}{(\sqrt{3})^2} = -\frac{1}{3}, \quad \theta_1 \cong 109.5^\circ$$

$\theta_1 = 1.23$ radians

Now vector $\vec{AG}$: $G = (0,1,1)$, $A = (1,0,0)$

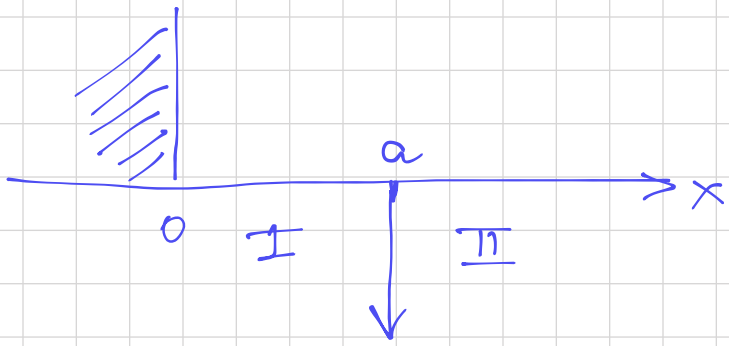
$\Rightarrow \vec{AG} = (-1,1,1)$

$\vec{AG} \cdot \vec{DF} = +1$. But $\vec{AG} \cdot \vec{DF} = (\sqrt{3})(\sqrt{3}) \cos\theta_2$

$$\Rightarrow \cos\theta_2 = +\frac{1}{3}, \quad \theta_2 \cong 70.5^\circ, \quad \theta_2 = 1.91 \text{ radians}$$

Evidently $\theta_1 + \theta_2 = 180^\circ$

$\theta_1 + \theta_2 = \pi$ radians.



(1) $$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \alpha\,\delta(x-a)\,\psi = -|E|\,\psi \quad \text{for } 0 \le x \le \infty$$

region I: $0 \le x \le a$

$$\psi_I(x) = A e^{\varkappa x} + B e^{-\varkappa x}, \quad \varkappa = \sqrt{\frac{2m|E|}{\hbar^2}}$$

At the infinite hard wall $\psi_I(x=0)=0$

$$\Rightarrow A + B = 0 \Rightarrow \psi_I(x) = 2A\,\mathrm{sh}(\varkappa x)$$

region II: $a \le x < \infty$

$$\psi_{II}(x) = C e^{-\varkappa x}$$

Boundary condition at $x=a$:

$$\psi_I(a-0) = \psi_{II}(x+0) \Rightarrow 2A\,\mathrm{sh}(\varkappa a) = C e^{-\varkappa a} \quad (2)$$

but derivatives are discontinuous!

To see this, integrate (1) over x around $x=a$ point:

$$-\frac{\hbar^2}{2m}\underbrace{\int_{a-\delta}^{a+\delta} dx\,\frac{d^2\psi}{dx^2}}_{\left.\frac{d\psi}{dx}\right|_{x=a+\delta} - \left.\frac{d\psi}{dx}\right|_{a-\delta}} - \alpha\underbrace{\int_{a-\delta}^{a+\delta} dx\,\delta(x-a)\,\psi(x)}_{\psi(a)} = -|E|\int_{a-\delta}^{a+\delta} dx\,\psi(x)$$

Take the limit $\delta \to 0$, then the right-hand side vanishes $\Rightarrow$ obtain $\left.\frac{d\psi}{dx}\right|_{a+0} - \left.\frac{d\psi}{dx}\right|_{a-0} = -\frac{2m\alpha}{\hbar^2}\psi(a)$

This gives $$-\varkappa C e^{-\varkappa a} - 2A\varkappa\,\mathrm{ch}(\varkappa a) = -\frac{2m\alpha}{\hbar^2}\mathrm{sh}(\varkappa a)\cdot 2A$$

$$C e^{-\varkappa a} + 2A\,\mathrm{ch}(\varkappa a) = \frac{2m\alpha}{\varkappa\hbar^2}\mathrm{sh}(\varkappa a)\cdot 2A \quad (3)$$

Using eq. (2) $C e^{-\varkappa a} = 2A\,\mathrm{sh}(\varkappa a)$

$\Rightarrow$ eq. (3) simplifies $\left(\frac{2m\alpha}{\hbar^2\varkappa} - 1\right) 2A\,\mathrm{sh}(\varkappa a) = 2A\,\mathrm{ch}(\varkappa a)$

$$\frac{2m\alpha}{\hbar^2\varkappa} - 1 = \frac{\mathrm{ch}(\varkappa a)}{\mathrm{sh}(\varkappa a)}$$

let $z = \varkappa a \Rightarrow \frac{2m\alpha a}{\hbar^2 z} = 1 + \frac{\mathrm{ch}\, z}{\mathrm{sh}\, z}$

<u>Limits:</u>

For $z \to \infty$ (which happens at finite $\varkappa$ and $a \to \infty$)

$$\frac{\mathrm{ch}\, z}{\mathrm{sh}\, z} \to 1 \Rightarrow \frac{2m\alpha a}{\hbar^2 z} \approx 2 \Rightarrow z = \frac{m\alpha a}{\hbar^2}$$

$$\Rightarrow \varkappa = \frac{m\alpha}{\hbar^2}.$$

this is standard result for the bound state energy in δ-function potential.

For $z \to 0$ (which happens when $\varkappa \to 0$, that is when bound state disappears)

$$\frac{\mathrm{ch}\, z}{\mathrm{sh}\, z} \to \frac{1}{z} \Rightarrow \frac{2m\alpha a}{\hbar^2 z} \approx 1 + \frac{1}{z}$$

$$z + 1 = \frac{2m\alpha a}{\hbar^2} \Rightarrow z = \frac{2m\alpha a}{\hbar^2} - 1 \geq 0$$

$$\Rightarrow a \geq a_{min} = \frac{\hbar^2}{2m\alpha}.$$

$\ast$ For $a > a_{min}$ the bound state exist.

For $a < a_{min}$ no bound state: hard wall is too close to the δ-function and "squizes" the bound state away.

(a) S.E. $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \alpha\delta(x)\psi = E\psi$

Integrate both sides over $x \in (-d, d)$:

$$-\frac{\hbar^2}{2m}\int_{-d}^{d}\frac{d^2\psi(x)}{dx^2}dx - \alpha\int_{-d}^{d}\delta(x)\psi(x)\,dx = E\int_{-d}^{d}\psi(x)dx.$$

Take the limit $d \to +0$:

$$-\frac{\hbar^2}{2m}\left(\frac{d\psi(+0)}{dx} - \frac{d\psi(-0)}{dx}\right) - \alpha\psi(0) = 0$$

$\Rightarrow$ boundary condition on $\frac{d\psi}{dx}$: $\frac{d\psi(+0)}{dx} - \frac{d\psi(-0)}{dx} = -\frac{2m\alpha}{\hbar^2}\psi(0)$

boundary condition of wave function: $\psi(+0) = \psi(-0)$ <u>always</u>

(b) Find solutions of S.E. for $x<0$ and $x>0$, and then match them using boundary conditions from (a):

$$\left.\begin{aligned}\psi(x<0) &= A e^{+\varkappa x}\\ \psi(x>0) &= B e^{-\varkappa x}\end{aligned}\right\}$$

looking for bound state, $E_0<0$

$\varkappa^2 = \frac{2m|E_0|}{\hbar^2}$

1. $\psi(+0) = A = \psi(-0) = B$

2. $\frac{d\psi(+0)}{dx} = -B\varkappa e^{-\varkappa x}\Big|_{x=0} = -B\varkappa$

$\frac{d\psi(-0)}{dx} = A\varkappa e^{\varkappa x}\Big|_{x=0} = A\varkappa$

$\Rightarrow -B\varkappa - A\varkappa = -\frac{2m\alpha}{\hbar^2}A \Rightarrow \boxed{\varkappa = \frac{m\alpha}{\hbar^2}}$

Now normalize:

$$1 = \int_{-\infty}^{\infty}dx\,|\psi(x)|^2 = 2\int_0^{\infty}dx\,A^2 e^{-2\varkappa x} = \frac{2A^2}{2\varkappa}$$

$\Rightarrow A = \sqrt{\varkappa}$

$$\psi(x) = \sqrt{\varkappa}\,e^{-\varkappa|x|}, \quad \varkappa = \frac{m\alpha}{\hbar^2}, \quad E_0 = -\frac{\hbar^2}{2m}\varkappa^2 = -\frac{m\alpha^2}{2\hbar^2}$$

(c) Sudden perturbation : $V(x) = -\alpha\delta(x) \rightarrow \tilde{V}(x) = -\tilde{\alpha}\,\delta(x)$

For this new potential the bound state energy is $\tilde{E}_0 = -\dfrac{m\tilde{\alpha}^2}{2\hbar^2}$, wave function $\tilde{\psi}(x) = \sqrt{\tilde{\varkappa}}\, e^{-\tilde{\varkappa}|x|}$

$$\tilde{\varkappa} = \frac{m\tilde{\alpha}}{\hbar^2}$$

(d) Initially $(t<0)$ $\psi(x) = \sqrt{\varkappa}\, e^{-\varkappa|x|}$, $\varkappa = \dfrac{m\alpha}{\hbar^2}$

At $t>0$, the bound state is given by part (c).

Probability to remain in the bound state

$$P_0 = |\langle\tilde{\psi}|\psi\rangle|^2 = \left(\int_{-\infty}^{\infty} dx\, \tilde{\psi}^*(x)\,\psi(x)\right)^2$$

The integral required:

$$J = \int_{-\infty}^{\infty} dx\, \sqrt{\tilde{\varkappa}}\, e^{-\tilde{\varkappa}|x|}\, \sqrt{\varkappa}\, e^{-\varkappa|x|} =$$

$$= \sqrt{\varkappa\tilde{\varkappa}} \cdot 2 \cdot \int_0^{\infty} dx\, e^{-(\varkappa+\tilde{\varkappa})x} = \frac{2\sqrt{\varkappa\tilde{\varkappa}}}{\varkappa+\tilde{\varkappa}}$$

$$P_0 = |J|^2 = \frac{4\varkappa\tilde{\varkappa}}{(\varkappa+\tilde{\varkappa})^2} = \frac{4\alpha\tilde{\alpha}}{(\alpha+\tilde{\alpha})^2}.$$

Quick simple limits to check the result:

1. $\tilde{\alpha} = \alpha \Rightarrow P_0 = \dfrac{4\alpha^2}{(2\alpha)^2} = 1$ <u>OK</u>

2. $\tilde{\alpha} = 0$ or $\alpha = 0 \Rightarrow P_0 = 0$ as then no bound state is present.